NEP 2.0

Ahmednagar Jilha Maratha Vidya Prasarak Samaj's

New Arts, Commerce, and Science College Ahmednagar (Autonomous)

(Affiliated to Savitribai Phule Pune University, Pune)

National Education Policy (NEP)
Choice Based Credit System (CBCS)

B. Sc. Computer Science-II (Major)

Implemented from

Academic Year 2025-26

Ahmednagar Jilha Maratha Vidya Prasarak Samaj's

New Arts, Commerce and Science College, Ahmednagar (Autonomous)

Board of Studies in Computer Science

Sr. No.	Name	Designation
1.	Prof. Arun Gangarde	**Chairman**
2.	Prof. P.U. Patil	**Member**
3.	Prof. M.B. Bhingare	**Member**
4.	Prof. M.B. Gobare	**Member**
5.	Prof. B.M. Danave	**Member**
6.	Prof. D.V. Jagdale	**Member**
7.	Prof. Vaibhav Ekshinge	**Member**
8.	Dr. M.N. Shelar	**Hon.Vice-Chancellors nominee**
9.	Dr. Shankar Mali	**AcademicCouncil Nominee**
10.	Dr. Chandrashekhar Patil	**AcademicCouncil Nominee**
11.	Mr. U.C. Temkar	**Industrial Expert**
12.	Mr. Prajwal Bhalsing	**Post Graduate Meritorious Alumnus**

B. Sc. Programme Framework: Credit Distribution

Level / Difficulty	Sem	Subject-1 (Selected as Major)						Subject-2		Subject-3		(SEC)	GE/OE		IKS	AEC	VEC	CC	Total
		T		P				T	P	P	T	P	T	P					
Certificate 4.5 / 100	I	02		02				02	02	02	02	-	02		02	02	02	02	22
	II	02		02				02	02	02	02	02	-	02	--	02	02	02	22
		Credits Related to Major						Selected as Minor											
		Core		Elective		VSC	FP / OJT/ CEP/RP												
		T	P	T	P	P	P	T	P	-		P	T	P	-	-	-	-	-
Diploma 5.0 / 200	III	04	02	--		02	02	02	02	-		02	02		-	02	-	02	22
	IV	04	02	--		02	02	02	02	-		02		02	--	02	-	02	22
Degree 5.5 /300	V	06	04	02	02	2	2	02	-	-		-	-		02	-	-	-	22
	VI	06	04	02	02	2	4	02	-	-		-	-		-	-	-	-	22
Total		24	16	04	04	08	10	10	08	04	04	06	08		04	08	04	08	132
6.0/400 Honours	VII	08	06	02	02	-	RM-04												22
	VIII	08	06	02	02		OJT-04												22
6.0/400 Honours with Research	VII	06	04	02	02		RM-04 RM-04												22
	VIII	06	04	02	02		RM-08												22
Total		40/36	28/24	08	08	08	18/26	10	08	04	04	06	04	04	04	08	04	08	176

B.Sc. Programme Framework: Course Distribution

Level / Difficulty	Sem	Subject-1 (Selected as Major)						Subject-2		Subject-3		(SEC)	GE/OE		IKS	AEC	VEC	CC	Total
		T		P				T	P	P	T	P	T	P					
Certificate 4.5 / 100	I	01		01				01	01	01	01	-	01		01	01	01	01	11
	II	01		01				01	01	01	01	01	-	01	--	01	01	01	11
		Credits Related to Major						Selected as Minor											
		Core		Elective		VSC	FP / OJT/ CEP/RP												
		T	P	T	P	P	P	T	P	-		P	T	P	-	-	-	-	-
Diploma 5.0 / 200	III	02	01	--		01	FP-01	01	01	-		01	01		-	01	-	01	11
	IV	02	01	--		01	CEP-01	01	01	-		01		01	--	01	-	01	11
Degree 5.5 /300	V	03	02	01	01	01	FP-01	01	-	-		-	-		01	-	-	-	11
	VI	03	02	01	01	01	OJT-01	01	-	-		-	-		-	-	-	-	10
Total		12	08	02	02	04	04			02	02	03	04		02	04	02	04	65
6.0/400 Honours	VII	03	03	01	01	-	RM-01												09
	VIII	03	03	01	01		OJT-01												09
6.0/400 Honours with Research	VII	02	02	01	01		RM-01 RM-01												08
	VIII	02	02	01	01		RM-01												07
Total		18/16	14/12	04	04	04	06/07	06	04	02	02	03	04		02	04	02	04	83/80

B. Sc. Computer Science: Credit and Course Distribution in Brackets

Level / Difficulty	Sem	Subject-1							Total
		T		P					
4.5	I	02 (01)		02 (01)					04(02)
	II	02 (01)		02 (01)					04(02)
		Credits Related to Major						IKS	
		Core		Elective		VSC	FP / OJT/ CEP		
		T	P	T	P	P	P	T	
5.0	III	04(02)	02(01)	--		02(01)	FP-02(01)		10(05)
	IV	04(02)	02(01)	--		02(01)	CEP-02(01)		10(05)
5.5	V	06(03)	04(02)	02(01)	02(01)	02(01)	FP-02(01)	02(01)	20 (10)
	VI	06(03)	04(02)	02(01)	02(01)	02(01)	OJT-04(01)		20(09)
Total		**12**	**08**	**(02)**	**(02)**	**04**	**04**	**(01)**	**33**
6.0	VII	03	03	(01)	(01)	-	RM-04(01)		22(09)
	VIII	03	03	(01)	(01)		OJT-04(01)		22(09)
6.0	VII	(02)	(02)	(01)	(01)		RM-04(01) RP-04(01)		22(08)
	VIII	(02)	(02)	(01)	(01)		RM-08(01)		22(07)
		18/16	14/12	04	04	04	06/07	(01)	51/48

Programme Framework (Courses and Credits): S.Y. B. Sc. Computer Science

Sr. No.	Year	Semester	Level	Course Type	Course Code	Title	Credits
1.	**II**	**III**	**5.0**	**DSC-05**	**BSCS231T**	**Data Structure and Algorithms**	**02**
2.	**II**	**III**	**5.0**	**DSC-06**	**BSCS232T**	**Web Technologies**	**02**
3.	**II**	**III**	**5.0**	**DSC-07**	**BSCS233P**	**Lab course based on CS231T(Data Struture and Algorithms)**	**02**
4.	**II**	**III**	**5.0**	**VSC-01**	**BSCS234P**	**Lab course based on CS232T (Web Technologies)**	**02**
5.	**II**	**III**	**5.0**	**FP-01**	**BSCS235P**	**Field Project**	**02**
6.	**II**	**IV**	**5.0**	**DSC-08**	**BSCS241T**	**Object oriented concept using c++**	**02**
7.	**II**	**IV**	**5.0**	**DSC-09**	**BSCS242T**	**Core Python**	**02**
8.	**II**	**IV**	**5.0**	**DSC-10**	**BSCS243P**	**Lab course based on CS241T(Object oriented concept using c++)**	**02**
9.	**II**	**IV**	**5.0**	**VSC-02**	**BSCS243T**	**Lab course based on CS242T(Core Python)**	**02**
10.	**II**	**IV**	**5.0**	**CEP-01**	**BSCS245P**	**Community Engagement Project**	**02**

Ahmednagar Jilha Maratha Vidya Prasarak Samaj's

New Arts, Commerce and Science College, Ahmednagar

(Autonomous)

Syllabus

B. Sc. -II (Computer Science)

Title of the Course: Data Structure and Algorithms								
Year: II				Semester: III				
Course Type	Course Code	Credit Distribution		Credits	Allotted Hours	Allotted Marks		
		Theory	Practical			CIE	ESE	Total
DSC-05	BSCS231T	02	00	02	30	15	35	50

Learning Objectives:

1. To provide the knowledge of basic data structures and their implementations.
2. To understand importance of data structures in context of writing efficient programs.
3. To develop skills to apply appropriate data structures in problem solving.

Course Outcomes (Cos):

1. Differentiate primitive and non-primitive structures.
2. To use well-organized data structures in solving various problems.
3. To differentiate the usage of various structures in problem solution.
4. Implementing data structures in various applications.

Detailed Syllabus:

Unit I: Introduction **(03)**

1.1 Concept
1.2 Datatype, Data Object Abstract Datatype (ADT)
1.3 Data structure Classification
1.4 Need of data structure
1.5 Application of data structure
1.6 Algorithm: Definition, characteristics, algorithm analysis (Time and space complexity)
1.7 Asymptotic notation

Unit II: Sorting Terminologies **(08)**

2.1 Array: Definition, 1-D, 2-D, multidimensional array
2.2 Sorting: Definition, types of sorting (Internal, External, In Place)
2.3 Lower bound on comparison based sorting
    2.3.1 Insertion sort
    2.3.2 Bubble sort
    2.3.3 Selection sort
2.4 Divide and conquer strategy
    2.4.1 Quick sort
    2.4.2 Merge sort
2.5 Non comparison based sorting
    2.5.1 Counting sort
    2.5.2 Radix sort

Unit III: Linear Dynamic Data Structure-I (Linked list) **(10)**

3.1 List as data structure, difference with array (linked list)
3.2 Static/Dynamic implementation of linked list
3.3 Types of linked list
    3.3.1 Singly linked list

3.3.2 Doubly linked list
3.3.3 Circular linked list
3.4 Operations on linked list
3.4.1 Create, insert, delete, search, reverse, concatenate, merge, traverse
3.5 Time complexity
3.6 Application of linked list
3.6.1 Polynomial representation
3.7 Generalized link list (GLL)
3.7.1 Concepts, representation, polynomial representation using GLL

Unit IV: Linear Dynamic Data Structure-II (Stack) (04)

4.1 Definition of stack
4.2 Operations on stack
4.2.1 init(), push(), pop(), isempty(), isfull(), peek()
4.3 Time complexity
4.4 Implementation: Static and Dynamic
4.5 Applications of stack
4.5.1 Function call & recursion
4.5.2 String reversal, palindrome checking
4.5.3 Expression types
4.5.3.1 Infix, Prefix, Postfix
4.5.3.2 Expression conversion and evaluation

Unit V: Linear Dynamic Data Structure-III (Queue) (03)

4.1 Introduction
4.2 Operations
4.2.1 Enqueue(), dequeue(), isempty(), isfull(), peek()
4.3 Implementation (Static & Dynamic)
4.4 Types of Queue
4.4.1 Linear Queue
4.4.2 Circular Queue
4.4.3 Priority Queue
4.4.4 Double ended Queue
4.5 Applications
4.5.1 CPU scheduling in multiprogramming environment.
4.5.2 Round Robin Algorithm

Unit VI: Non-linear Data structure (Tree) (02)

6.1 Introduction
6.2 Types of Tree
6.2.1 Binary Tree
6.2.2 Skewed Tree
6.2.3 Strictly binary Tree
6.2.4 Full binary Tree
6.3 Representation

Suggested Readings:

1. Data Structures, Horowitz, Sahani, Galgotia publication

2. Fundamentals of Computer Algorithms, Ellis Horowitz, Sartaj Sahni and Sanguthevar Rajasekara, Galgotia Pub. 2001 ed.

3. Data Structures Through C in Depth, S.K.Srivastava and Deepali Srivastava, BPB Publication

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Syllabus

B. Sc. -II (Computer Science)

Title of the Course:Web Technologies								
Year: II				Semester: III				
Course Type	Course Code	Credit Distribution		Credits	Allotted Hours	Allotted Marks		
		Theory	Practical			CIE	ESE	Total
DSC-06	BSCS232T	02	00	02	30	15	35	50

Learning Objectives:

1. Describe different web technologies and application development issues and trends.
2. Distinguish between server-side and client-side web technologies.
3. Apply CSS types for different web pages and technologies of World Wide Web as a platform.
4. Validate web form fields using JavaScript.

Course Outcomes (Cos)

1. To introduce the fundamentals of Internet, and the principles of web designing
2. To construct basic websites using HTML and Cascading Style Sheets.
3. To build dynamic web pages with validation using Java Script objects and by applying different event handling mechanisms.
4. To develop modern interactive web applications using HTML,CSS,Java Script.
5. Detailed Syllabus:

Unit I Introduction to web **(06)**

1.1 Internet
- 1.1.1 Definition
- 1.1.2 History of Internet
- 1.1.3 World Wide Web
- 1.1.4 Web client, Web server
- 1.1.5 Basic internal Protocol
- 1.1.6 Client server architecture (Two tier, Multitier)
- 1.1.7 HTTP Request & HTTP Response

1.2 URL

Unit II: Hypertext Markup Language **(10)**

2.1 Introduction

2.2 Basic structure of HTML Document
- 2.2.1 HTML editors
- 2. 2.2 Markup Tags: heading, paragraphs, formatting, line-breaks
- 2.2.3 HTML color, link, image
- 2.2.4 Working with text, lists, tables, frames, blocks
- 2.2.5 Working with Hyperlinks
- 2.2.6 HTML Images
- 2.2.7 HTML forms

2.2.8 Form Layout
2.2.9 HTML Iframe
2.3 Web Design
2.3.1 Concepts of effective web design,
2.3.2 Web design issues including
2.3.3 Bandwidth and Cache,
2.3.4 Display resolution,
2.3.5 Look and Feel of the Website
2.3.6 Page Layout
2.3.7 Designing effective navigation.

Unit III:Cascading Style Sheets (07)

3.1 Introduction, Syntax & selectors
3.2 Ways to insert CSS
3.3 CSS properties
3.3.1 Color
3.3.2 Border
3.3.3 Margin
3.3.4 Padding
3.3.5 Height & Width
3.3.6 Text-formatting, list, tables
3 . 5 Layout
3.5.1 The display property
3 . 5 . 2 The positionary property

Unit IV Basic Of JavaScript (07)

4.1 Introduction

4.2 Datatypes of JavaScript
4.3 Control Structure of JavaScript
4.4 Examples of JavaScript

4.5. JavaScript DOM
4.5.1 Java Script Functions
4.5.2 Working with events
3 5.3 JS Popup Boxes ()
4.5.4 JavaScript objects
4.5.5 Java Script HTML Dom
4.5.6 Forms and validation

Suggested Readings/Material:

1. HTML 5 Black Book, Covers CSS 3, JavaScript, XML, XHTML, AJAX, PHP and jQuery, 2ed Paperback – 1 January 2016 by DT Editorial Services (Author)
2. Web Technologies: HTML, JAVASCRIPT, PHP, JAVA, JSP, ASP.NET, XML and Ajax, Black Book: HTML, Javascript, PHP, Java, Jsp, XML and Ajax, Black Book Paperback – 1 January 2009 by Kogent Learning Solutions Inc. (Author)
3. JavaScript | e Paperback – 1 April 2021
by Archana Devi (Author), Chellaprabha (Author), Shriram K. Vasudevan (Author), Sunandhini Muralidharan (Author)

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Syllabus

B. Sc. -II (Computer Science)

Title of the Course: Lab course based on CS 231T(Data Structure and Algorithms)								
Year: II				**Semester: III**				
Course Type	Course Code	Credit Distribution		Credits	Allotted Hours	Allotted Marks		
		Theory	Practical			CIE	ESE	Total
DSC-07	BSCS233P	00	02	02	60	15	35	50

Learning Objectives:

1. focus on hands-on experience, enabling students to design, implement, and analyze algorithms and data structures, choose appropriate data structures for problems, and understand time and space complexity.

Course Outcomes (Cos):

1. Understanding the concept of dynamic memory management.
2. Understanding basic data structure such as array, queue, linked list, stack.
3. To develop the methods and procedures for software development that can scale up for large system and can be used consistently to produce high quality software.

Detailed Syllabus:

Assignment 1:
Implementation of sorting algorithms:
Bubble sort, Insertion Sort & Selection sort
Assignment 2:
Divide and Conquer Strategy:
i) Implementation of Merge sort ii) Implementation Quick sort
Assignment 3:
Non-comparison based sort:
i) Implementation of Count sort ii) Implementation of Radix sort
Assignment 4
i) Structures-
structure with array, structure with function, structure with pointer
ii) Pointer-
pointer with array, pointer with structure, pointer to pointer
iii) Use of typedef, dynamic memory allocation
Linear singly link list:
i) Static implementation of linear singly linked list
ii) Dynamic implementation of linear singly linked list
iii) Create, insert, delete, display, search, count, Reverse
Assignment 5
Application of linear singly linked list:
i) Merge two linked list
ii) Concatenation of two linked list
iii) Polynomial addition of linked list

Assignment 6

Linear doubly linked list:

i) Dynamic implementation of linear singly linked list

Create, insert, delete, display, search, count, Reverse

Assignment 7

Circular linked list:

i) Dynamic implementation of circular singly linked list

ii) Dynamic implementation of circular doubly linked list Basic operations:

Create, display, delete, insert, delete

Assignment 8

Stack:

Static and dynamic implementation of stack to perform basic operations:

init(), push(), pop(),peek(), isfull(), isempty()

Applications of Stack

Assignment 9

Queue:

Implementation of (Static and Dynamic) of Queue to perform basic operations:

Create, Insert, Delete, Display

Implementation of circular queue

Suggested Readings/Material:

1. Laboratory handbook prepared by the college.

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New Arts, Commerce and Science College, Ahmednagar

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Syllabus

B. Sc. -II (Computer Science)

Title of the Course: : Lab course based on CS232T(WebTechnologies)								
Year: II				Semester: III				
Course Type	Course Code	Credit Distribution		Credits	Allotted Hours	Allotted Marks		
		Theory	Practical			CIE	ESE	Total
VSC-01	BSCS234P	00	02	02	60	15	35	50

Learning Objectives:

1. Understanding the relationship of HTML, CSS & JavaScript.
2. Create cascading stylesheet(CSS) for device and browser integration
3. Utilize the concept of JavaScript.

Course Outcomes (Cos) :

1. To introduce the fundamentals of Internet, and the principles of web designing
2. To construct basic websites using HTML and Cascading Style Sheets.
3. To build dynamic web pages with validation using Java Script objects and by applying different event handling mechanisms.
4. To develop modern interactive web applications using HTML,CSS,Java Script.

Detailed Syllabus: Example

Assignment 1

Creating Sample HTML page using Basic tags of HTML. (Text formatting).

Assignment 2

HTML Programming using list, tables hyper link, frame,

Assignment 3

HTML forms designing Using from tag.

Assignment 4

Introduction to CSS & its type. Implementation of CSS element selector.

Assignment 5

Implementation of box model CSS, (Margin padding, border)

Assignment 6.

Implementation of Display and Position property &Basic transition in CSS

Assignment 7

Implementation of Control structure in Java script.

Assignment 8

Implementation of array &its object in Java script.

Assignment 9

Implementation of function in java script.

Assignment 10

Implementation of pop-up box in java script.

Assignment 11

Implementation of different events in java script.

Assignment 12

DOM manipulation and form validation using Java Script.

Suggested Readings/Material:

Laboratory handbook prepared by the college.

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Syllabus

B. Sc. -II (Computer Science)

Title of the Course: Field Project								
Year: II				**Semester: III**				
Course Type	Course Code	Credit Distribution		Credits	Allotted Hours	Allotted Marks		
		Theory	Practical			CIE	ESE	Total
FP-01	BSCS235P	00	02	02	60	15	35	50

1. The department shall conduct the awareness programme/session on how to complete the Field Project (Major related) at the beginning of the semester.
2. The department forms batches and appoints guides/mentors for each batch.
3. The department includes a session of 04 hours/batch in the timetable to guide and evaluate students every week.
4. Students need to prepare a detailed field project report and submit it to the department.
5. The Field Project will be evaluated based on presentation and report as per guidelines.
6. Field Project Guidelines Approved by Academic Council: https://newartscollege.ac.in/wp-content/uploads/2024/07/Field_Project_Guidelines1.pdf

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Syllabus

B. Sc. -II (Computer Science)

Title of the Course: Co-curricular Course							
Year: II			**Semester: III**				
Course Type	Course Code	Credit Distribution Course/Activity Based	Credits	Allotted Hours	Allotted Marks		
					CIE	ESE	Total
CC	**CC03**	02	02	60	15	35	50

Sr. No.	**Course Code**	**Year**	**Semester**	**Humanities**	**Commerce**	**Science**	**Credits**
1.	CC03 & CC04	II	III & IV	(Any one of the following) * 1. NSS/NCC 2. Publication of an article, essay, or poem in the college magazine, newspapers, or any other recognized magazine. 3. Small project work showcasing achievements of India in different fields. 4. Participation in cultural or sports activities at university/state/national level. 5. Participation in community works related to national integration, environment, Human rights, disaster management, etc. with recognized GO or NGO or College. 6. Completion of certificate course organised by any department of the college 7. SWAYAM Course 8. Book Review as per guidelines prescribed by the college. 9. College representation in Dance/Music /Theatre/Visual Arts Competitions at the university/state/national level. 10. MKCL iLike Courses			02
* Each Co-curricular course/activity shall have a capping of 02 credits except for NSS/NCC and SWAYAM Courses. i.e., if a student selects any one CC activity/programme in the III semester, he/she shall not be allowed to select the same activity in the IV semester. However, students are allowed to complete 02 different SWAYAM Courses in the III and IV semesters respectively. # Details guidelines for acquiring 04 credits of CC for NCC cadets and NSS volunteers will be shared separately.							

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Syllabus

B. Sc. -II (Computer Science)

Title of the Course: Object Oriented Concepts Using C++								
Year: II				**Semester: IV**				
Course Type	Course Code	Credit Distribution		Credits	Allotted Hours	Allotted Marks		
		Theory	Practical			CIE	ESE	Total
DSC-08	BSCS241T	02	00	02	32	15	35	50

Learning Objectives:

1. Aim to understand and implement key principles like encapsulation, inheritance, polymorphism, and abstraction to design modular, reusable, and maintainable code.

Course Outcomes (Cos)

1. To learn the fundamental concepts and methodologies which are essential to build C++ programs.
2. Apply the concepts of object oriented programming.
3. Apply virtual and pure virtual functions and complex programming.

Detailed Syllabus:

Unit I: Introduction of C++ **(05)**

1.1 Ethics of Object Oriented Programming
- 1.1.1 Object oriented programming paradigm.
- 1.1.2 Basic concepts of object oriented programming. (objects, classes, Data abstraction, Data encapsulation, inheritance, polymorphism, Dynamic binding, message passing)
- 1.1.3 Benefits of OOPs
- 1.1.4 Applications of OOPs

1.2 What is C++?
1.3 Application of C++
1.4 Structure of C++ program
1.5 Simple C++ program
1.6 Managing Console I/O Operations
- 1.6.1Streams of C++
- 1.6.2 Stream Class of C++
- 1.6.3 Unformatted I/O Operations
- 1.6.4 Formatted I/O Operations
- 1.6.5 Managing output with Manipulators

Unit II: Tokens, Expressions of C++ **(02)**

2.1 Tokens
2.2 Keywords
2.3 Identifiers
2.4 Basic data types
2.5 User defined data types
2.6 Derived data types

2.7 Symbolic constants
2.8 Dynamic initialization of variables
2.9 Reference variables
2.10 Operators in C++
2.11 Scope resolution operators
2.12 Memory dereferencing operators
2.13 Memory management operators
2.14 Manipulators
2.15 Type cast operators

Unit III: Functions in C++ **(04)**
3.1 Function prototyping
3.2 Call by reference
3.3 Return by reference
3.4 Inline function
3.5 Function overloading
3.6 Friend function

Unit IV: Class & Objects in C++ **(02)**
4.1 Specifying a class and creating object of class
4.2 Defining member function
4.3 A C++ program with class
4.4 Memory allocation of object
4.5 Array of objects
4.6 Static data member
4.7 Static member function

Unit V: Constructor and Destructor **(06)**
5.1 Definition of constructor
5.2 Types of constructor
5.3 Destructor
5.4 Operator overloading
    5.4.1 Definition, rules for overloading operators
    5.4.2 Unary operator overloading
    5.4.3 Binary operator overloading

Unit VI: Inheritance **(10)**
6.1 Definition, visibility modifier (private, public, protected)
6.2 Types of inheritance
    6.2.1 Single inheritance
    6.2.2 Multilevel inheritance
    6.2.3 Multiple inheritance
    6.2.4 Hierarchical inheritance
    6.2.5 Hybrid inheritance
6.3 Abstract classes
6.4 Virtual base classes
6.5 Virtual function
6.6 Rules for virtual function
6.7 Pure virtual function
6.8 This pointer

Unit VI: Working with Files **(03)**

6.1 Classes for file stream operations
6.2 Opening and closing files
6.3 File pointers & their manipulations
6.4 Sequential I/P & O/P operations

Suggested Readings/Material:

1. Object-oriented programming with C++ by E.Balagurusamy, 2nd Edition, TMH.

2. C++: The Complete Reference- Schildt, McGraw-Hill Education (India)

3. Mastering C++ - Venugopal, McGraw-Hill Education (India)

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Syllabus

B. Sc. -II (Computer Science)

Title of the Course: Core Python								
Year: II				Semester: IV				
Course Type	Course Code	Credit Distribution		Credits	Allotted Hours	Allotted Marks		
		Theory	Practical			CIE	ESE	Total
DSC-09	BSCS242T	02	00	02	30	15	35	50

Learning Objectives:

1. Master the fundamentals of writing Python scripts.
2. Learn core Python scripting elements such as variables and flow control structures.
3. Discover how to work with lists and sequence data.
4. Write Python functions to facilitate code reuse.
5. Use Python to read and write files.

Course Outcomes (Cos)

1. Acquire knowledge and programming skills in python
2. Students will learn the concepts like Basics of Python programming
3. Decision Making and Functions in Python.

Detailed Syllabus

Unit I **(07)**

Introduction to python interpreter

1.1 Introduction to Python Programming Language
1.2 History of python
1.3 features of python programming
1.4 Applications areas of python programming
1.5 Running Simple Python program
1.6 Basics of Python
- 1.6.1 Standard data types
- 1.6.2 Variables and Constants,
- 1.6.3 Python identifiers and reserved words
- 1.6.4 Lines and indentation
- 1.6.5 Multi-line Statements
- 1.6.6 Comments
- 1.6.7 Input/output with print and input

1.7 Operator in Python
- 1.7.1 AssignOperator
- 1.7.2 Arithmetic Operator
- 1.7.3 Logical Operator
- 1.7.4 Bitwise Operator
- 1.7.5 Membership Operator
- 1.7.6 Identity Operator
- 1.7.7 Precedence of operators
- 1.7.8 Type conversion

1.8 Writing Simple program
1.8.1 Examples

Unit II Control Statement **(07)**

2.1 ConditionalStatements
2.1.1 if Statement
2.1.2 if else Statement
2.1.3 if elif else Statement
2.1.3 nested if-else Statement
2.1.4 Assert Statement
2.2 Loop
2.2.1forloop
2.2.2while loop
2.3 nested Loop Control Statements
2.3.1 break
2.3.2 continue
2.3.3 pass
2.4 Strings
2.4.1 declaration ofString
2.4.2 String manipulation
2.4.3 Special operations on String
2.4.4 Escapecharacters
2.4.5 Stringformattingoperator
2.4.6 Raw String and Unicode strings
2.4.7 Built-in String methods

Unit III Lists, Functions tuples, dictionaries, Sets **(08)**

3.1 Python Lists
3.1.1 Defining List
3.1.2 creating and accessing elements
3.1.3 updating & deleting lists
3.1.4traversing a List and range function
3.1.5 reverse Built-in List Operators
3.1.6 Basic list operation
3.1.6.1 Concatenation, Repetition, InOperator
3.1.7 Built-inList functions and methods

3.2 Function
3.2.1 Function Definition
3.2.2 Function Calls
3.2.3 Type Conversion Functions
3.2.4 Bulit in functions
3.2.5 Math Functions
3.2.6 Composition
3.2.7 Adding New Functions
3.2.8 Flow of Execution
3.2.9 Parameters andArguments
3.2.10 Variables an Parameters
3.2.11 Boolean Functions
3.2.12 functional programming tools
3.2.12.1 Recursion
3.2.12.2 map ()
3.2.12.3 filter ()

3.2.12.4 reduce ()

3.3 Tuples

3.3.1 Accessing values in Tuples
3.3.2 Tuple Assignment
3.3.3 Tuples as return values
3.3.4 Variable-length argument tuples
3.3.4.1 Basic tuples operations
3.3.4.2 Repetition
3.3.4.3 in Operator
3.3.4.4Iterating tuple value
3.3.4 Built-in tuplefunctions
3.3.4.1 indexing
3.3.4.2 slicing
3.3.4.3 matrices

3.4 Dictionary

3.4.1 DefiningDictionary
3.4.2 creating and accessing elements
3.4.3 updating & deleting Dictionary element
3.4.4 Properties of Dictionary keys
3.4.5 Operations in Dictionary
3.4.6 Built-In Dictionary Functions
3.4.7Built-inDictionaryMethods
3.4.8 Variable-length argument Dictionary
3.4.9 Zip function

3.5 Set

3.5.1 Definition
3.5.2 Operations of set (Adding, Union, intersection)
3.5.3 3.5.3working with sets

Unit IV Modules, File Handling and Exception handling (08)

4.1 Modules

4.1.2 Importing module
4.1.3 Creating &exploringmodules
4.1.4 Standard in buildModules
4.1.4.1 Math module
4.1.4.2 Random module
4.1.4.3 Date and Time module

4.2 Packages

4.2.1 Importingpackage
4.2.2 creating package
4.2.3 examples

4.3 File Handling

4.3.1 Creating files
4.3.2 Operations on files (open, close, read, write)
4.3.3 File objectattributes
4.3.4 file positions
4.3.5 isting Files in a Directory
4.3.6 Testing File Types
4.3.7 Removing files and directories

4.3.8 copying and renaming files
4.3.9 splitting path names
4.3.10 creating and moving directories

Suggested Readings/Material:

1.An Introduction to Computer Science using Python 3 by Jason Montojo, Jennifer Campbell, Paul Gries, The pragmatic bookshelf-2013
2 .James Payne, “Beginning Python: Using Python and Python 3.1,Wrox Publication
3 .Introduction to Problem Solving with Python by E balguruswamy,TMH publication2016

Ahmednagar Jilha Maratha Vidya Prasarak Samaj's

New Arts, Commerce and Science College, Ahmednagar

(Autonomous)

Syllabus

B. Sc. -II (Computer Science)

Title of the Course: Lab course based on CS 241T (Object Oriented Concepts using C++)								
Year: II				**Semester: IV**				
Course Type	Course Code	Credit Distribution		Credits	Allotted Hours	Allotted Marks		
		Theory	Practical			CIE	ESE	Total
DSC-10	BSCS243P	00	02	02	60	15	35	50

Learning Objectives:

Implement OOP principles, create classes and objects, use inheritance and polymorphism, and develop applications with file I/O and exception handling.

Course Outcomes (Cos):

1. Able to differentiate structure oriented programming and object oriented programming.
2. Able to understand and apply various object oriented features.
3. Able to know concepts in operator overloading, function overloading & polymorphism.
4. Able to write, compile and debug programs in C++ language.
5. Design programs involving constructors, destructors.
6. Able to reuse of code using inheritance.
7. To implement the concept of files, templates and exceptions.
8. To write diversified solutions using C++ language.

Detailed Syllabus:

Assignment 1

Structure of C++ program, Header file, comments, main functions, C++ Program template, The standard output stream, cascading of I/O operation.

Assignment 2

Class, object, public, private, protected, defining member function. member function defined inside the class, member function defined outside the class, array of object manipulators.

Assignment 3

function, inline function, friend function, static class member and static member function. static data members, static member function.

Assignment 4

Function overloading way to overload a function, default arguments, sample programs and Examples.

Assignment 5

constructors and destructors: constructors, types of constructors, default constructor, copy constructor, dynamic constructors, destructor dynamic memory allocation operator new operator.

Assignment 6.

Operator overloading, function overloading, function overriding

Assignment 7

Inheritance, implementation of inheritance in C++, types of Inheritance,

Assignment 8

Virtual function & polymorphism: Compile time polymorphism, runtime polymorphism, examples virtual functions in derived class, pure virtual class & abstract class, examples

Assignment 9

Working with files: File, stream, opening file, closing file, text file

Suggested Readings/Material:

Laboratory handbook prepared by the college.

Ahmednagar Jilha Maratha Vidya Prasarak Samaj's

New Arts, Commerce and Science College, Ahmednagar

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Syllabus

B. Sc. -II (Computer Science)

Title of the Course: : Lab course based on CS242T(Core Python)								
Year: II				Semester: IV				
Course Type	Course Code	Credit Distribution		Credits	Allotted Hours	Allotted Marks		
		Theory	Practical			CIE	ESE	Total
VSC-02	BSCS244P	00	02	02	60	15	35	50

Learning Objectives:

1. To define the structure and components of a Python program. □
2. To learn how to write loops and decision statements in Python. □
3. To learn how to write functions and pass arguments in Python. □
4. To learn how to build and package Python modules for reusability. □

Course Outcomes (Cos)

1. Students will be able to understand why Python is a useful scripting language for developers. □
2. Students will learn how to design and program Python applications. □

Assignments 1:

Introduction:

Introduction to python programming language and basic concept
like comment, Indentation. basic programming using print and input function.

Assignments 2:

Operator in python

Demonstration of arithmetic operator, relational operator,
logical operator, bitwise operator membership operator and identity operator

Assignments 3:

Control statement

Conditional Statement
Demonstration of Conditional Statements if statement, if else
Statement,
if elif else Statement, nested if-else Statement, Assert Statement

Assignments 4:

Control statement Loop

Demonstration of looping statement while, for and nested loop and
loop control Statement break, continue and pass

Assignments 5:

List

Demonstration of List and various operation on the list

Assignments 6:

unction

Demonstration of function and writing user define functions

Assignments 7:

Function

Anonymous (lambda function)
Demonstration of Anonymous (lambda function) and writing
Anonymous (lambda function)

Assignments 8:

Function

Boolean function
Demonstration of Boolean function and writing Boolean
function

Assignments 9:

Function

Function programming tools
Demonstration of function programming tools and programming using
map(),
Filter(), reduce method
Function
Department of Computer Science
Recursion
Demonstration of recursion and programming using recursion

Assignments 10:

Tuple

Demonstration of Tuple and various operation on the tuple

Assignments 11 Tuple:

Variable-length argument tuples
Demonstration of Variable-length argument Function and writing function
with Variable-length argument

Assignments 12:

Dictionary

Demonstration of Dictionary and various operation on the
Dictionary

Assignments 13:

Dictionary

Variable-length argument Dictionary
Demonstration of Variable-length argument Function and writing function

with Variable-length argument

Assignments 14:

Dictionary

Zip function

Demonstration of zip() function and writing programs using zip function

Assignments 15:

Set

Demonstration of Set and various operation on the Set

Assignments 16:

Standard Modules

Demonstration of Standard Modules (Date and Time, Math and Random) and programming usng the methods from them

Assignments 17:

File Handling

Demonstration of file in python and writing program for file manipulation

Suggested Readings:

Laboratory handbook prepared by the college.

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New Arts, Commerce and Science College, Ahmednagar

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Syllabus

B. Sc. -II (Computer Science)

Title of the Course: Community Engagment Program								
Year: II				**Semester: IV**				
Course Type	Course Code	Credit Distribution		Credits	Allotted Hours	Allotted Marks		
		Theory	Practical			CIE	ESE	Total
CEP-01	BSCS245P	00	02	02	60	15	35	50

7. The department shall conduct the awareness programme/session on how to complete the Community Engagement Project (Major related) at the beginning of the semester.
8. The department forms batches and appoints guides/mentors for each batch.
9. The department includes a session of 04 hours/batch in the timetable to guide and evaluate students every week.
10. Students need to prepare a detailed Community Engagement project report and submit it to the department.
11. The Community Engagement Project will be evaluated based on presentation and report as per guidelines.
12. Community Engagement Project Guidelines Approved by Academic Council: https://newartscollege.ac.in/wp-content/uploads/2024/07/CE_Project_Guidelines1.pdf

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New Arts, Commerce and Science College, Ahmednagar

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Syllabus

B. Sc. -II (Computer Science)

Title of the Course: Co-curricular Course							
Year: II			**Semester: IV**				
Course Type	Course Code	Credit Distribution	Credits	Allotted Hours	Allotted Marks		
		Course/Activity Based			CIE	ESE	Total
CC	**CC04**	02	02	60	15	35	50

Sr. No.	**Course Code**	**Year**	**Semester**	**Humanities**	**Commerce**	**Science**	**Credits**
1.	CC03 & CC04	II	III & IV	(Any one of the following) * 11. NSS/NCC 12. Publication of an article, essay, or poem in the college magazine, newspapers, or any other recognized magazine. 13. Small project work showcasing achievements of India in different fields. 14. Participation in cultural or sports activities at university/state/national level. 15. Participation in community works related to national integration, environment, Human rights, disaster management, etc. with recognized GO or NGO or College. 16. Completion of certificate course organised by any department of the college 17. SWAYAM Course 18. Book Review as per guidelines prescribed by the college. 19. College representation in Dance/Music /Theatre/Visual Arts Competitions at the university/state/national level. 20. MKCL iLike Courses			02
* Each Co-curricular course/activity shall have a capping of 02 credits except for NSS/NCC and SWAYAM Courses. i.e., if a student selects any one CC activity/programme in the III semester, he/she shall not be allowed to select the same activity in the IV semester. However, students are allowed to complete 02 different SWAYAM Courses in the III and IV semesters respectively. # Details guidelines for acquiring 04 credits of CC for NCC cadets and NSS volunteers will be shared separately.							